

Weekly Progress Report

Name: Devika Polavarapu

Domain: Data Science and Machine Learning

Date of submission: May 15, 2026

Week Ending: 04

I. Overview:

This week, the primary focus was on finalizing machine learning projects and improving understanding of model optimization, result analysis, and workflow management. Efforts were also made to strengthen implementation skills and overall project documentation.

II. Achievements:

1. USC_TIA Familiarization:

- Improved understanding of project workflow and reporting process.
- Successfully managed project documentation and submission activities.

2. Python Project Contributions:

Name of the project:- Traffic Forecasting, Crop Production Prediction

- Finalized traffic forecasting project and analyzed prediction accuracy using evaluation metrics.
- Improved crop production prediction project through preprocessing refinement and model comparison.
- Worked on better understanding of feature impact and model behavior.

3. Learning Python:

- Strengthened practical knowledge in pandas, numpy, scikit-learn, matplotlib, and seaborn.
- Improved confidence in implementing complete machine learning workflows independently.

III. Challenges:

1. USC_TIA Integration:

- Faced challenges in organizing reports and maintaining proper workflow consistency.
- Resolved by carefully reviewing guidelines and maintaining structured documentation.

2. Python Project Complexity:

- Faced difficulty in improving model performance and interpreting prediction results.
- Addressed by comparing models and analyzing evaluation outputs.

IV. Learning Resources:

1. USC_TIA Documentation:

- Utilized USC_TIA guidelines and references for proper project submission.
- Referred to online tutorials for resolving workflow-related doubts.

2. Python Learning Resources:

- Explored machine learning documentation and practical examples for better understanding.
- Continued learning through hands-on experimentation with datasets and models.

V. Next Week's Goals:

1. USC_TIA Enhancement:

- Continue improving workflow management and documentation practices.

2. Python Project Development:

- Explore additional machine learning concepts and advanced model techniques.
- Focus on improving visualization and presentation of project results.

VI. Additional Comments:

"This week helped in strengthening practical machine learning knowledge and improving confidence in handling complete project implementation and evaluation."